

Steel Manufacturing Incident Analysis and Prediction

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ABSTRACT

Employees of the U.S. steel manufacturing industry encounter dangerous working environments often resulting from limited visibility, hazardous proximity situations between heavy equipment and pedestrian workers and the dynamic nature of manufacturing tasks. Working conditions typical of steel manufacturing environments include increased amounts of repetitive work tasks, elevated temperature, noisy surroundings and an overall rugged work environment. These conditions tend to cultivate conditional and behavioral hazards that increase the probability of employees experiencing an incident in the form of an injury, illness or fatality. In an attempt to proactively identify hazardous situations and conditions, details from safety incident data can be analyzed to identify predictor variables of future of incidents in steel manufacturing environments.

The objective of this research is to utilize statistical modeling to identify specific variables that have high correlations of an unsafe event or condition within a steel manufacturing facility. Safety lagging indicator data (injuries, illnesses and fatalities) from an active steel manufacturing facility were input into multiple statistical predictive models to better understand how individual safety metrics can predict incidents. Results indicate weather, task performance, moving equipment, location, defective equipment, and personal responsibility all greatly increase the probability of an injury, illness or fatality occurring within a steel manufacturing environment. The contribution of this research lies in the presented framework for predicting variables that significantly impact lagging indicators as well as scientific evaluation of the framework.

Keywords: *Lagging indicators, safety performance metrics, steel manufacturing safety*

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1. Introduction

Members of the U.S. steel manufacturing industry continue to experience a significant number of injuries, illnesses and fatalities (Reimink, 2016). The combination of intricate technology and physical labor create a complicated challenge for safety managers in steel manufacturing (Verma, et al. 2014). Like other larger industrial sectors, steel manufacturing has implemented pro-active safety program elements to improve safety performance of employees (Cambraia, et al. 2010; Lander, et al., 2011). For example, many steel manufacturing companies implement hazard identification programs in an attempt to identify and mitigate hazards before an injury, illness or fatality occurs (Basso, et al., 2004). Based on results of the identified hazards, mitigation strategies are created and implemented (Mohammadfam, et al., 2011). Although these pro-active safety programs are successful, a knowledge gap exists between analyzing incident data and understanding high impact variables for incidents, injuries and illnesses.

The objective of this research is to find elements from near miss reports that have a significant correlation with injuries, illnesses and fatalities in an active steel manufacturing environment. Statistical prediction models are implemented to identify correlations of multiple variables derived from safety data from a steel manufacturing facility collected over eight months. The Binary Logistic Regression Method was implemented for this study. The dataset contained approximately 2,300 incidents and was divided into 63 variables. The goal of this research is to provide insight into elements that are common to injuries, illnesses and fatalities in steel manufacturing in an effort to enhance safety performance in this type of work environment.

2. Literature Review

The following section reviews the steel manufacturing industry and associated hazards, the current state of safety

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in steel manufacturing and statistical predictive models. A research needs statement is derived from results of the review.

2.1 U.S. Steel Manufacturing Industry

The U.S. steel manufacturing industry produced 6,382,000 tons of steel in October 2016 (TE, 2016). The U.S. steel industry operators more than 100 steel producing facilities which produce 87 million tons of steel valued at \$75 billion in 2014 (AISI, 2016). This industry is a segment of primary metals manufacturing which employed approximately 372,800 people in August 2016 (BLS, 2016) and accounted for 1.4% of primary energy consumption in 2006 in the U.S. (U.S. EIA, 2012; Zhang, et al., 2010). Almost 69% of all steel manufactured in North America is recycled each year (AISI, 2016). After production, steel manufacturing companies ship 42% of their product to construction sites and 27% to automotive manufacturing facilities (AISI, 2016).

2.2 Steel Manufacturing Incidents and Hazards

The U.S. steel manufacturing industry is one of the most dangerous when compared to other industrial sectors in the U.S. Statistics show that Steel foundries has an incident rate of 9.1% of total nonfatal occupational injury and illness cases (BSL, 2015b). The primary metal manufacturing industry experienced 28 workplace fatalities in 2014 which increased from the 17 fatalities experienced in 2013 (BLS, 2015b). The main cause of these fatalities was fires, explosions and contact with equipment or objects (BLS, 2015a). The primary metal manufacturing industry also experiences a significant number of non-fatal injuries. The industry recorded 20,000 non-fatal injuries in 2014 and 19,900 in 2013 (BLS, 2015b). 20,540 illnesses resulting from workplace hazards were experienced by primary metal manufacturing personnel in 2014 (BLS, 2015b). This number was slightly lower from the 21,320 illnesses experienced by this industrial sectors in 2013 (BLS, 2015b). For both years, the main contributing cause to the illnesses was exposure to noise (BLS, 2015b). These injury, illness and fatality statistics indicate that much improvement is required in steel manufacturing safety performance.

Steel manufacturing employees experience multiple hazards including exposure to high temperatures (Dhar et al. 2006), exposure to chemicals (Safty, et al., 2008), ergonomic problems (Han, et al., 2007) and noise exposure (Ologe, et al., 2006). The implementation of specific measures for preventing workplace injury and illness in the steel manufacturing industry depends largely on the recognition of principle hazards (ILO, 2005). One study identified several risks that steel manufacturing workers accept as part of workplace conditions including noise and heat exposure (Nordlof, et al., 2015). Several safety climate surveys of steel manufacturing have revealed the major underlying problems were inadequate health and safety procedures, pressures for production and employee experience (Adl, et al., 2011; Baek, et al., 2008). Furthermore, management com-

mitment, organizational industrial relations and the political environment impact safety hazard awareness through the implementation of safety policy (Karkoszka & Szewieczek, 2007). Although hazard identification provides a useful method for mitigating hazards, the impact of specific hazards categories on injuries, illnesses and fatalities has not been quantified.

2.3 Predicting Safety Performance

Due to the perceived value in safety performance prediction, researchers have attempted to estimate safety performance of steel manufacturing employees from various perspectives. Correlations between incident causes and specific types of incidents were identified for a steel manufacturing facility (Verma, et al., 2014). For example, the study found that unsafe acts done by subcontractors are more frequent than injury cases (Verma, et al., 2014). One steel manufacturing safety study claimed that patterns of employee norms predicted perceptions of work environment safety and at-risk behavior (Watson, et al., 2005). The study also found an employee group's belief in management's safety values predicted at risk behaviors for steel manufacturing workers (Watson, et al., 2005). Safety hazards, safety culture and production pressure were found to significantly influence employee's attitudes with regard to safety in a steel manufacturing facility (Brown, et al., 2000). Minimal research has been conducted to quantify specific variables that contribute to injuries, illnesses and fatalities in the steel manufacturing industry.

2.4 Research Needs Statement

The steel manufacturing industry needs a scientific approach to identify and predict specific variables that increase the probability of an unsafe event or condition within a steel manufacturing facility. This approach can be achieved through statistical analysis and prediction modeling of historical steel manufacturing incidents. By understanding the impact variables of steel manufacturing incidents, critical elements to each event can be identified. This information can equip industry leaders to recognize critical causation attributes and mitigate before an injury, illness or fatality occurs.

3. Methodology

The purpose of this research is to identify and analyze variables from safety incident reports that significantly impact the probability of an unsafe event or condition within a steel manufacturing facility. The safety incidents used were from an active steel manufacturing facility in the U.S. that specializes in carbon coil and plate markets. Approximately 2,300 reported incidents from employees at the steel manufacturing facility between January 2010 and August 2016 were input into statistical predictive models.

3.1 Variable Description

A large steel manufacturing company in the U.S. provided researchers access to a safety incident database reported by

employees and analyzed by safety managers at the company. Variables retrieved from employee safety incident logs were organized by safety managers into company-specific safety categories. To perform predictive statistical models, variable categorizes were defined as either independent or dependent variables. The dependent variables included all metrics associated with the outcome of a reported incident such as an injury, illness or fatality. These incident types were referred to as response variables because they were hypothesized to result from the independent variables. Nine groups of incident types were extracted from the original dataset and divided into the following two categories: 1) safety lagging indicators and 2) safety leading indicators. Unlike lagging indicators, safety leading indicators are safety metrics that occur before an injury, illness or fatality occurs and are generally a measure of performance processes and work activities (Hallowell, et al., 2013).

For this research, incident type was the selected dependent variable. The dependent variable included nine categories which are defined in Table 1.

Category	Description
Lagging Indicators	
OSHA Recordable	An injury or illness that results in death, days away from work, restricted work, medical treatment beyond first aid or loss of consciousness (OSHA, 2001).
Lost time	A specific type of OSHA recordable in which an employee is unable to return to work or is assigned restricted work on the day or shift following the incident (OSHA, 2011).
First aid	Any incident that requires stopping work but does not require a trained medical professional for assistance (OSHA, 2016).
Property damage	An incident that results in destruction to real or personal property.
Environmental Incident	Incident resulting in unsafe working conditions or hazardous environmental conditions.
Fire	Incident resulting in explosion and/or fires.
Leading Indicators	
Restricted duty/modified work	Occurs when an employer keeps the employee from performing one or more of the routine functions of his/her job (OSHA, 2006).
Policy violation	Incident caused by an employee failing to follow an OSHA regulation or company specific safety policy (Hallowell, et al., 2013).
Near miss	An unplanned event or unsafe condition that has the potential for injury or illness to people, or damage to property, or the environment (Hallowell, et al., 2013).

Table 1. Description of dependent (response) variables.

Due to the limitation of Binary Logit Model, the dependent variable has to be dichotomous. The variable incident types were changed into two new dichotomous variables named “lagging indicator” and “leading indicator.” The lagging indicator was selected as the dependent variable in the model in attempt to identify what variables impact these events. These dependent, or response variables can be statistically explained by the combination of independent variables. All dependent variables are quantitative and are calculated through count-

ing individual events. Although the categorization system is susceptible to human error, categorizes were assigned by one individual safety manager within the company following a consistent decision criteria. Figure 1 shows a screenshot of some of the variables recorded when incidents occurred in the steel manufacturing facility.

The independent variables implemented included aspects of the reported incident not including the incident type. Table 2 provides a definition of all 16 independent variables assessed. It is hypothesized that these independent, or predictor variables can statistically explain the occurrence of dependent variables or incident types for a steel manufacturing facility. Each of the variables listed in Table 2 is a binary variable.

Category	Description
Summer Indicator	Incident occurred in June, July or August
Task Performed Indicator 1	Incident occurred during operating
Task Performed Indicator 2	Incident occurred during driving
Moving Equipment Indicator 1	Incident occurred during operating crane
Moving Equipment Indicator 2	Incident occurred during operating truck
Moving Equipment Indicator 3	Incident occurred during operating forklift
Moving Equipment Indicator 4	Incident occurred during operating trailer
Mobile Equipment Indicator	Incident involved mobile equipment
Location Indicator 1	Incident occurred in Roll Shop
Location Indicator 2	Incident occurred in Coil Yard or Disposition
Location Indicator 3	Incident occurred in Cut to Length (CTL) shop
Location Indicator 4	Incident occurred in West Gate
Location Indicator 5	Incident occurred in Water System
Location Indicator 6	Incident occurred in Melt Shop
Preliminary Cause Indicator 1	Incident caused by defective equipment
Preliminary Cause Indicator 2	Incident caused by personal responsibility

Table 2. Description of independent (predictor) variables.

Based on the information of the original dataset, the dataset was categorized into 62 independent variables. Variables can be divided into the following six categories: 1) incident time of day indicator; 2) month of incident indicator; 3) incident details indicator; 4) task performed indicator; 5) mobile equipment indicator; and 6) location indicator. Both incident details indicator and task performed indicator were categorized by key word search from the original text description of the incident reporter. The incident details category includes descriptions in which a specific task was performed when incident occurred. These variables were categorized into the following five independent variables: 1) operating; 2) removing; 3) loading; 4) driving; and 5) dumping. The task performed indicator category represents moving equipment that was operated when incident occurred. These incidents were further categorized as: 1) crane; 2) truck; 3) ladle; 4) forklift; and 5) trailer. Finally, 16 independent variables were selected to test the model.

IncidentID	IncidentDate	AreaDesc	DeptDesc	SpecificLocation	IncidentDetails
1	9/13/08 9:30 PM	EMF	Melt Shop	Old Carbon Silo	Driver stepped out of his truck and rolled his ankle over on
2	9/14/08 8:00 AM	Melt Shop	Melt Shop	Scrap inspection stand	The driver parked his truck and went to get his paper work
3	12/2/08 12:00 PM	Mobile Maint	Maintenance	Northport Auto Supply picking u	Incident Description
4	12/6/08 7:45 AM	EMF/LMF/Refi	Melt Shop	Behind Flood Wall, West side of	He was operating the Genie 135 man lift. He was in the pro
5	12/11/08 12:15 PM	NULL	NULL	Refueling Station-Rolling Mill	He was moving the manlift to the refueling station by the Rc
6	12/11/08 11:32 AM	EMF	Melt Shop	ER#13	Incident Description Install electric box at ER#13, both cont
7	12/14/08 1:30 PM	Caster	Melt Shop	Tundish Lifting Jig Stand	He was in the process of picking up the tundish lifting jig wit

Figure 1. Sample database screenshot.

3.2 Binary Logit Analysis

Multi-level models are especially useful when individual variables are subcategories within larger units and it is thought that both individual-level and group-level elements impact the dependent variable (Lau, 2007). Mixed-effects regression models have become increasingly popular for clustered data-sets (Goldstein, 1995). Due to the organization and metrics recorded for the steel manufacturing safety incident database analyzed in this research, the Binary Logit Model was one statistical prediction model selected for analysis. The probability, denoted $\Pr(Y)$, is assumed to be determined by a set of independent variables ($X_1, X_2, \dots, X_j$), and a corresponding set of parameters ($\beta_0, \beta_1, \beta_2, \dots, \beta_j$). It is specified as a linear function of the independent variables by the following equation (Hainen 2016):

$$Y_i = \beta X_i + \epsilon_i \quad i = 1, 2, \dots, n \quad \text{Equation 1}$$

where, X_i is the independent variables (lagging indicators). β is a vector of estimable parameters and ϵ_i is an error term. The logit distribution constrains the estimated probabilities between zero and one. The estimated probability is:

$$\Pr(Y) = \frac{e^{\beta}}{1 + e^{\beta}} \quad \text{Equation 2}$$

In the logit model, the logistic cumulative density function is (Young & Liesman, 2007):

$$\beta = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_j X_j \quad \text{Equation 3}$$

If Y is the binary response variable, then the following statements are valid:

- 1) $Y_i = 1$ if the trait is present in observation (lagging indicator) i
- 2) $Y_i = 0$ if the trait is not present in observation i

Additionally, $Y = 1$ meaning that x_i equals one if the trait is present in the observation (i.e., summer indicator is true) and $y_i = 0$ otherwise. The statistical software program Nlogit with the logistic procedure was used to estimate the maximum likelihood probability function. The Binary Model was executed with the safety incident dataset from the active steel manufacturing facility. Results of the model are provided in the subsequent section.

4. Results

Results from the completed Binary Logit Model are presented in this section. Detailed outcomes of each model and potential impacts to safety in the steel manufacturing environment are discussed.

4.1 Results of Binary Logit Model

For this research, all independent variables that entered the model were considered at the significant level of 0.05. No other independent variable met the correlation coefficient value of 0.10 for entry into the model. The log-likelihood of this model is -734 and all parameters were included if their t -statistic was greater than 1.96. The total number of observations included

Variable Description	Estimated Parameter	t-statistics
Summer Indicator	0.415	2.78***
Incident Details Indicator 1	0.470	2.46***
Incident Details Indicator 2	1.130	2.39***
Task Performed Indicator 1	-0.457	-1.94***
Task Performed Indicator 2	0.813	2.00***
Task Performed Indicator 3	1.158	2.28***
Task Performed Indicator 4	0.729	1.74**
Mobile Equipment Indicator	0.001	3.93***
Location Indicator 1	1.279	1.87**
Location Indicator 2	1.315	2.63***
Location Indicator 3	1.492	5.15***
Location Indicator 4	-2.243	-2.55***
Location Indicator 5	-1.222	-2.39***
Location Indicator 6	-0.619	-1.88**
Preliminary Cause Indicator 1	0.247	3.41***
Preliminary Cause Indicator 2	-0.247	-3.42***
Sample size, n	2,300	
Log-likelihood	-734	

***0.01, **0.05, *0.10

Table 3. Binary Logit estimation for potential probability of an incident occurrence.

Variable Description	Marginal effect
Summer Indicator	0.0918
Incident Details Indicator 1	0.1024
Incident Details Indicator 2	0.2216
Task Performed Indicator 1	-0.1039
Task Performed Indicator 2	0.1677
Task Performed Indicator 3	0.2260
Task Performed Indicator 4	0.1519
Mobile Equipment Indicator	0.0002
Location Indicator 1	0.2856
Location Indicator 2	0.2938
Location Indicator 3	0.3332
Location Indicator 4	-0.5011
Location Indicator 5	-0.2730
Location Indicator 6	-0.1383
Preliminary Cause Indicator 1	0.0551
Preliminary Cause Indicator 2	-0.0553

Table 4. Marginal effect of the potential probability of an incident occurrence.

in the model was 1,153. Numeric outputs of the Binary Logit Model concerning the potential probability of an incident occurrence are provided in Table 3 and corresponding marginal effects are presented in Table 4.

Outputs from the Binary Logit Model identified injuries were associated with employees of the steel manufacturing company working in the summer (include June, July or August) with t -statistics of 2.78 and marginal effect of 0.0918. This steel manufacturing firm is located in Southeastern region of U.S. which experiences elevated temperatures during the summer. One possible explanation for this higher possibility is that severe weather is potentially more impactful to employees than normally weather. Employee may be distracted and fatigued due to the high temperature. According to the outputs of the Binary Logit Model, incident details indicators related to operating and driving with t -statistics of 2.46 and 2.39 represent a large correlation of having an incident. This statistic verifies the current proximity problem in the manufacturing environment between pieces of equipment and pedestrian employees.

The output parameter of task performed indicators which

includes cranes provided a t -statistic of -1.94 which indicates cranes have a negative correlations of having incidents. One can hypothesize that recent increased safety attention of cranes can project an increased awareness of activities involving cranes in steel manufacturing environments. Other outputs of the Binary Logit Model show that workers who operator trucks, trucks and trailers and forklifts have a relatively large correlations of having an incident with a t -statistic of 2.00, 2.28 and 1.74, respectively. These pieces of equipment are typical violators of hazardous proximity situations between pieces of equipment and pedestrian employees. Each of these pieces of equipment have limited visibility around the equipment footprint which adds to the overall proximity hazard. It is also important to note that employees who are required to be near mobile equipment have a high t -statistic of 3.93 demonstrating that pedestrian employees near mobile equipment have a high positive correlations to injuries. This finding is consistent with past research which identified hazardous proximity situations for pedestrian employees in manufacturing environments (Horberry, et al., 2004).

The location indicators analysis found that employees working at the west gate, water system or melt shop have negative correlation to have an incident with t -statistics of -2.55, -2.39 and -1.88, respectively. The west gate was the main entrance used for shipping products and receiving materials from transportation delivery trucks (the marginal effects of location indicator in Table 4 show that the west gate had the largest impact). The area denoted as water system or melt shop is an environment where scrap metal is melted into liquid steel and placed into molds for creating structural steel elements. These work environments are clearly marked as no pedestrian areas which has helped eliminate risk and safety problems in these areas. On the contrary, employees located in the roll shop, coil yard or cut-to-length shop have positive correlations to have an incidents with t -statistics of 1.87, 2.63 and 5.15 respectively. These areas require pedestrian workers and thus are subject to hazardous proximity situations between equipment and workers-on-foot. These specific findings for this data set are useful for this specific steel manufacturing plant, but this same approach can be applied to other steel manufacturing plants as well as manufacturing plants in general. This system provides a custom-tailored analysis to understand risky locations at other manufacturing environments.

After assessing the preliminary cause indicator, defective equipment with a t -statistic of 3.41 has a positive correlation of resulting in an incident. This analysis suggests that defective equipment in the steel manufacturing facility were more likely to result in an incident. Because newer equipment typically includes more safety features, one can understand the reason for this high probability. Furthermore, this may help provide feedback for upgrades to equipment and then post-upgrade review.

Lastly, the personal responsibility indicator projected a t -statistic of -3.42 which indicates a negative correlation of resulting in an incident. For this specific facility, it seems unsafe

employee behavior was not common and thus did not result in many incidents. This indicates the observed personal safety behaviors had a negative correlation of an incident. Although subjective, an inverse correlation can be hypothesized between the less injuries and a higher quality safety culture within this steel manufacturing company.

5. Conclusion

This research implemented the Binary Logit Model to analyze the correlations of lagging indicators within a steel manufacturing facility as well as identified specific impact variables for incidents. The model was selected to analyze the factors that may have the correlations of potentially having an incident occurring from a reported incidents. Findings from the regression suggest that a positive correlations between incidents and summer months. Results also suggest that injuries has a positive correlations with pedestrian employees are near pieces of moving equipment. Mobile equipment including trucks, forklifts and truck and trailer combinations have a positive correlation of causing an incident. Overall, employees of the steel manufacturing company experience greater success with mitigating hazards from crane operation than other moving equipment operations.

Location indicators showing that employees who worked at locations with a high population of moving equipment were positive correlated with having an incident. Consequently, employees that work in environments free of moving heavy equipment have inverse correlations with incidents. For this specific steel manufacturing environment, defective equipment have high positive correlations between an incident and unsafe worker behavior was not prevalent in incidents.

Results of this research identify meaningful correlations between lagging indicators and incidents report elements. These findings provide insight into weather, task performance, moving equipment, location, defective equipment, and personal responsibility indicators. The contribution of this research lies in the presented framework for predicting variables that significantly impact lagging indicators as well as scientific evaluation of the framework. By identifying these impact variables and their connections to lagging indicators, safety managers can optimally direct mitigation efforts of their safety program. ■

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